

Bacterial Taurine Uptake and Desulfonation: A Two-Part TauABC–TauD Module for Sulfur Acquisition

A review-style synthesis for a molecular biology audience

1. Executive Summary

Bacteria that lack sufficient inorganic sulfate can scavenge sulfur from the abundant organosulfonate **taurine** (2-aminoethanesulfonate) using a compact, self-contained two-part module encoded by the **tauABCD** operon. The module works in an obligatory sequence: the **TauABC** ATP-binding cassette (ABC) importer first internalizes extracellular taurine, and the cytoplasmic **TauD** enzyme—an Fe(II)/2-oxoglutarate (2OG)-dependent dioxygenase—then oxidatively cleaves the carbon–sulfur bond, liberating **sulfite** (which enters sulfur assimilation) and **aminoacetaldehyde** as a carbon byproduct. The entire module is expressed only under sulfate or cysteine starvation, placing it firmly in the category of a scavenging system rather than a constitutive metabolic route.

The two halves of the module are mechanistically very different but functionally interdependent. Import specificity is governed by the periplasmic substrate-binding protein **TauA**, whose selectivity for taurine over other alkanesulfonates arises not from unique sulfonate coordination chemistry but from the *desolvation* thermodynamics of its binding pocket. Desulfonation is carried out by TauD, the **archetypal and best-studied member** of the entire Fe/2OG oxygenase superfamily. TauD was the enzyme in which the field's first high-valent **Fe(IV)=O (ferryl) intermediate**—"intermediate J"—was directly trapped and characterized, making it the reference system for understanding how these ubiquitous enzymes activate O₂ and abstract hydrogen atoms from unreactive C–H bonds. A hallmark of TauD's chemistry is that O₂ activation must be gated by bound substrate; when taurine is absent, the enzyme still activates O₂ and self-destructively hydroxylates an active-site tyrosine (Tyr73), a well-characterized failure mode that illuminates the coupling logic of the whole superfamily.

This review defines the boundaries of the module, lays out the best current mechanistic model for each half, situates the aerobic TauD route among mechanistically distinct alternatives (the sibling FMNH₂-dependent SsuABC/SsuD alkanesulfonate branch, and anaerobic taurine:pyruvate-aminotransferase/sulfo-lyase catabolism), and identifies the open questions—most prominently, the unresolved rebound-versus-alkoxide question of how the ferryl intermediate completes hydroxylation. Throughout, the evidence is strongest for *Escherichia coli* and *Pseudomonas putida*, and readers should be cautious about generalizing mechanistic detail across lineages that use taurine for different purposes.

2. Definition and Biological Boundaries

What the system is

The **bacterial taurine uptake and desulfonation module** is a two-step, sulfur-starvation-induced pathway that acquires taurine from the environment and extracts its sulfur:

1. **Uptake** — the **TauABC** ABC importer moves extracellular taurine across the cytoplasmic membrane.
2. **Desulfonation** — the cytoplasmic **TauD** dioxygenase cleaves the C–S bond, releasing sulfite and aminoacetaldehyde.

In *E. coli*, all four genes lie together in a ~1.8-kb region at 8.5 min on the chromosome and are transcribed from a single sulfate-starvation-regulated promoter ([PMID: 8808933](#)). The operon encodes a periplasmic binding protein (TauA), the ABC transporter components (TauB ATPase, TauC permease), and the 2OG-dependent dioxygenase TauD. The purpose of the module is **sulfur acquisition**: the released sulfite feeds into the sulfate-assimilation pathway to make cysteine, while the carbon skeleton (aminoacetaldehyde) is a byproduct.

What lies outside the boundary

Several neighboring processes are frequently mentioned alongside this module but are properly treated as separate systems:

- **Downstream aminoacetaldehyde metabolism** and **sulfite assimilation** — these consume the products of the module but are not part of it.

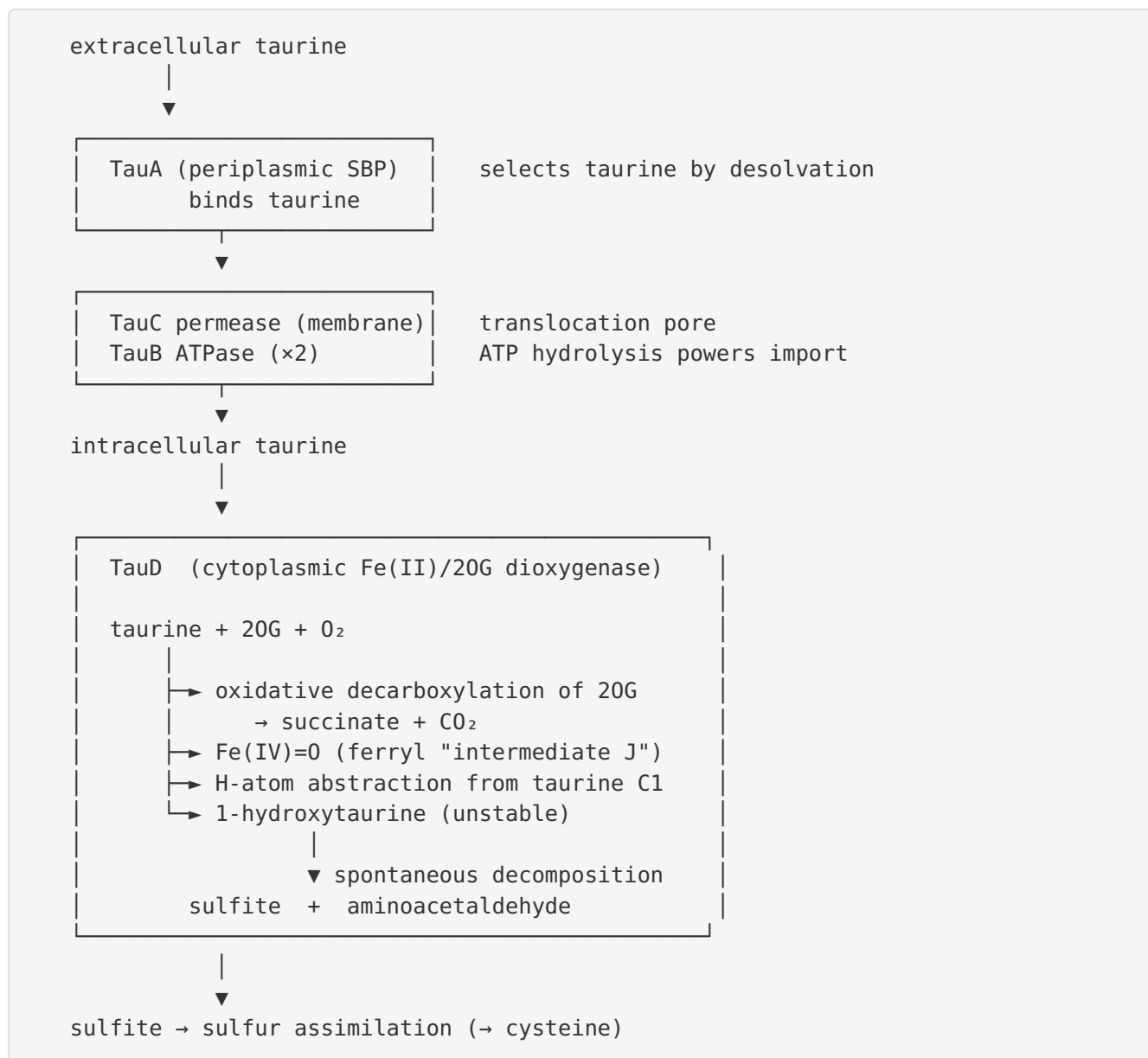
- **Sulfur-starvation signaling** (the CysB/Cbl regulatory circuit) — this controls *when* the module is expressed but is a regulatory layer, not a catalytic step (discussed in §5 because it defines the physiological state in which the module operates).
- **The paralogous SsuABC/SsuD alkanesulfonate branch** — a sibling system that imports and cleaves *other* alkanesulfonates using a completely different chemistry (an FMNH₂-dependent monooxygenase, SsuD, plus its reductase SsuE). It shares regulation and architecture with the tau module but is mechanistically distinct.
- **Anaerobic and catabolic taurine routes** — in some bacteria taurine is a carbon/nitrogen/energy source degraded by O₂-independent transaminase/sulfo-lyase chemistry, with no dioxygenase involved.

Competing definitions

The main definitional ambiguity in the literature is whether "taurine degradation" refers to the **sulfur-scavenging TauD route** or to the broader set of **taurine catabolic pathways** that use taurine as a carbon/nitrogen source. These are biochemically and physiologically distinct (see §5), and conflating them is a common source of confusion. This review adopts the narrow, sulfur-acquisition definition: TauABC import feeding TauD desulfonation.

3. Mechanistic Overview

The module operates as a strictly ordered, two-station assembly line:



Obligatory steps. Import must precede cleavage—TauD is cytoplasmic and acts only on internalized taurine, so TauABC supplies the substrate. Within TauD, the ordered binding of Fe(II), then 2OG (bidentate chelation), then taurine, then O₂ is required; O₂ reacts only after the substrate is in place. Oxidative decarboxylation of 2OG to succinate + CO₂ is coupled to formation of the ferryl intermediate, which abstracts a hydrogen atom from taurine's C1. The resulting 1-hydroxytaurine is intrinsically unstable and decomposes to sulfite and aminoacetaldehyde.

Conditional steps. Expression of the whole module is conditional on sulfur starvation (see §5). The choice of taurine versus other sulfonates depends on which importer/enzyme pair is available (tau vs ssu).

Accessory features. The self-hydroxylation "off-pathway" chemistry (§6) is an accessory, unproductive branch that occurs when coupling fails; it is not part of the productive cycle but is diagnostically important.

4. Major Molecular Players and Active Assemblies

4.1 The tauABCD operon and its induction

Finding F001. The taurine module is a two-part, sulfur-starvation-induced operon. In *E. coli*, tauABCD occupies ~1.8 kb at 8.5 min, is transcribed from a single sulfate-starvation-regulated promoter, and encodes a periplasmic binding protein (TauA), the ABC components (TauB ATPase, TauC permease), and the 2OG-dependent dioxygenase TauD. Disrupting the genes abolishes growth on taurine as a sole sulfur source (PMID: 8808933). The original genetic analysis stated plainly that *"the proteins encoded by tauABC constitute an uptake system for taurine and that the product of tauD is involved in the oxygenolytic release of sulfite from taurine."* Deletion analysis confirmed the division of labor and the conditional expression: the genes are *"expressed only under conditions of sulfate or cysteine starvation"* (PMID: 10781534).

4.2 TauABC: the ABC importer and its selectivity filter

Finding F002. TauABC is a canonical ABC importer. The periplasmic substrate-binding protein **TauA** captures taurine and delivers it to the TauBC membrane complex. Crystal structures and isothermal titration calorimetry (ITC) show that the sulfonate-coordinating residues are largely conserved between TauA and its paralog SsuA (the exception being Asp205, absent in SsuA). Strikingly, taurine selectivity is **not** explained by unique sulfonate contacts; instead, TauA binds taurine with a much lower enthalpic penalty than other alkanesulfonates, and molecular dynamics simulations attribute this to the *degree of hydration/desolvation of the binding site*: *"the different levels of hydration of the binding site contributed to the selectivity for taurine over the other alkanesulfonates"* (PMID: 31802112).

The tau and ssu systems have overlapping but distinct substrate ranges and are **not freely interchangeable**. Deletion analysis showed that *"mutants in which only formation of hybrid transporters was possible were unable to grow with sulfonates"* (PMID: 10781534), demonstrating that the components of the two importers do not mix-and-match into functional hybrids.

Component	Type	Location	Role
TauA	Periplasmic binding protein	Periplasm	Captures taurine; selectivity by desolvation
TauC	Integral membrane permease	Inner membrane	Translocation channel
TauB	ABC ATPase	Cytoplasmic face	ATP hydrolysis powers transport

4.3 TauD: the archetypal Fe(II)/2OG dioxygenase

Finding F003. TauD couples oxidative decarboxylation of 2OG (α -ketoglutarate) to hydroxylation of taurine's C1 carbon, producing an unstable 1-hydroxy intermediate that decomposes to sulfite and aminoacetaldehyde. Its central importance is historical as well as mechanistic: TauD was the enzyme in which the first high-valent iron intermediate in *any* 2OG dioxygenase was directly observed. Price and colleagues, using rapid kinetics, Mössbauer, and EPR, demonstrated that "*an oxidized Fe intermediate in the reaction of one of these enzymes, taurine/ α -ketoglutarate dioxygenase (TauD) from Escherichia coli, has been directly demonstrated by rapid kinetic and spectroscopic methods*" (PMID: 12809506). This species—a high-spin, formally Fe(IV) **ferryl** intermediate now universally called "**intermediate J**" or **TauD-J**—is the reactive oxidant.

The reaction stoichiometry and product identity are firmly established: TauD is "*the Fe(II)/ α -ketoglutarate-dependent taurine dioxygenase that catalyzes the hydroxylation of taurine yielding sulfite and aminoacetaldehyde*" (PMID: 22221834). That same study established that TauD from both *P. putida* and *E. coli* assembles as a **homotetramer**.

Substrate recognition is mediated by specific active-site residues. Kinetic and mutagenesis work "*confirmed the importance of His 70 and Arg 270 in binding the sulfonate moiety of taurine and indicated the participation of Asn 95 in recognizing the substrate amine group*" (PMID: 15751960). Thus the zwitterionic taurine molecule is anchored by its sulfonate (His70/Arg270) and its amine (Asn95), positioning C1 over the iron center.

4.4 The fold, the facial triad, and substrate geometry

Finding F006. The X-ray structure of *E. coli* TauD in complex with Fe(II), 2OG, and taurine revealed the canonical **double-stranded β -helix (DSBH / "jelly-roll") fold** shared across the superfamily, with the closest structural relative being clavamate synthase: "*the tertiary*

structure and fold of TauD are similar to those observed in other enzymes from the broad family of Fe(II)/alphaKG-dependent oxygenases, with closest structural similarity to clavamate synthase" (PMID: 11955067). The mononuclear iron is held by the hallmark **2-His-1-carboxylate facial triad** (His99, Asp101, His255). Systematic ligand mutagenesis confirmed these assignments: His99, "*coplanar with alphaKG and Fe(II), is unalterable in terms of maintaining an active enzyme,*" while D101E retains ~22% of kcat and His255 tolerates Gln/Glu substitutions (PMID: 17350690). The 2OG cofactor chelates iron in a bidentate manner, leaving a single coordination site open for O₂.

A key structural insight is how TauD discriminates substrate geometry. The active-site architecture selects taurine's **tetrahedral sulfonate anion** in preference to a planar carboxylate, distinguishing TauD from its close homolog TfdA (2,4-dichlorophenoxyacetate dioxygenase): the structure "*reveals how TauD selects a tetrahedral substrate anion in preference to the planar carboxylate selected by TfdA*" (PMID: 11955067). Indeed, the TauD coordinates provided the structural model for understanding TfdA-type enzymes before their own structures were available.

Feature	Detail
Fold	Double-stranded β -helix (jelly-roll); closest to clavamate synthase
Iron ligands	2-His-1-carboxylate facial triad: His99, Asp101, His255
Cofactor	2-oxoglutarate (bidentate Fe chelation) + O ₂
Substrate anchors	His70 + Arg270 (sulfonate); Asn95 (amine)
Substrate geometry	Tetrahedral sulfonate (vs planar carboxylate in TfdA)
Oligomeric state	Homotetramer
Reactive oxidant	High-spin Fe(IV)=O ferryl ("intermediate J")

4.5 The regulatory circuit that gates expression

Finding F004. The module is embedded in a **Cbl/CysB-controlled sulfonate regulon** whose signal is a metabolite of sulfate assimilation. Expression of tauABCD requires two LysR-type transcriptional regulators: **CysB** (the master sulfur regulator) and **Cbl** (an accessory, sulfonate-specific activator). A translational tauD'-lacZ fusion requires both: "*Expression of the tau and ssu genes requires the LysR-type transcriptional regulatory proteins CysB and Cbl*" (PMID: 11479697). Crucially, Cbl activation is switched off by a specific metabolite: Cbl-mediated

transcription "in vitro is abolished in the presence of an early metabolite of the sulphate assimilatory pathway, adenosine 5'-phosphosulphate (APS)" (PMID: 11918818). This identifies **APS—not sulfate itself—as the negative signaling co-factor** reporting sulfate sufficiency. When sulfate is plentiful, APS rises and shuts off the sulfonate regulon; when sulfate is scarce, APS falls and Cbl activates tau/ssu expression. The tau module thus sits alongside the parallel SsuABC transporter + SsuD monooxygenase branch within a shared, APS-gated regulon.

5. Evolutionary and Cell-Biological Variation

5.1 Multiple, mechanistically distinct routes to taurine desulfonation

Finding F005. The oxygenolytic TauABC/TauD route is only one of several ways bacteria handle taurine, and these routes differ fundamentally in chemistry, oxygen requirement, and physiological purpose.

- **The TauD route (aerobic, sulfur acquisition).** Strictly O₂-dependent; dedicated to obtaining sulfur. Uses a 2OG-dependent dioxygenase.
- **FMNH₂-dependent monooxygenase route (aerobic, other sulfonates).** Even within aerobic desulfonation, the chemistry is split by substrate: TauD (2OG-dependent) handles taurine, whereas the paralogous **SsuD** (an FMNH₂-dependent monooxygenase, with **SsuE** as its NAD(P)H-dependent FMN reductase) handles other alkanesulfonates. As summarized in a review of the field, "*Desulfonation of aromatic and aliphatic sulfonates as sulfur sources by aerobic bacteria is oxygen-dependent, carried out by the alpha-ketoglutarate-dependent taurine dioxygenase, or by one of several FMNH(2)-dependent monooxygenases*" (PMID: 10717312).
- **Anaerobic/catabolic transaminase route (O₂-independent, C/N/energy source).** In other lineages—such as *Rhodobacter capsulatus* and anaerobes like *Bilophila wadsworthia* and *Desulfonispora*—taurine is used as a carbon, nitrogen, or energy source through O₂-independent chemistry. Degradation initiates by **taurine:pyruvate aminotransferase (Tpa)**, followed by sulfoacetaldehyde acetyltransferase/sulfo-lyase, releasing sulfite/sulfoacetaldehyde with no dioxygenase involved. In *R. capsulatus*, an ORF "shows strong similarity to taurine:pyruvate aminotransferase (Tpa) from *Bilophila wadsworthia* catalyzing the initial transamination during anaerobic taurine degradation" (PMID: 11728723).

Route	O ₂ requirement	Key enzyme(s)	Purpose	Example organisms
TauABC/ TauD	Obligate aerobic	TauD (Fe/2OG dioxygenase)	Sulfur source	<i>E. coli</i> , <i>P. putida</i>
SsuABC/ SsuD	Obligate aerobic	SsuD (FMNH ₂ monooxygenase) + SsuE	Sulfur from alkanesulfonates	<i>E. coli</i>
Tpa/sulfo-lyase	O ₂ -independent	Taurine:pyruvate aminotransferase	C/N/energy source	<i>R. capsulatus</i> , <i>B. wadsworthia</i>

5.2 Conservation and the ancestral role of the TfdA-like family

TauD belongs to the **TfdA-like Fe/2OG oxygenase superfamily**, an ancient and vast enzyme family built on the DSBH fold and the 2-His-1-carboxylate facial triad. This chemistry is deeply conserved across all domains of life—the same facial-triad ferryl mechanism underlies hydroxylation reactions from bacterial antibiotic biosynthesis (clavamate synthase, TauD's closest structural relative) to eukaryotic collagen prolyl hydroxylases and DNA/RNA demethylases. Within this expansion, **TauD is the best-characterized representative** for understanding the ancestral hydroxylation chemistry, precisely because its ferryl intermediate was the first to be trapped and it has been the subject of the most detailed kinetic, spectroscopic, structural, and mutagenic dissection.

The relationship between TauD and TfdA (2,4-D dioxygenase) is instructive: the two are close homologs that diverged to accept substrates of different geometry (tetrahedral sulfonate vs planar carboxylate), and the TauD structure served as the template for modeling TfdA before its own structure was solved ([PMID: 11955067](#)). The paralogous relationship between the tau and ssu importers and between TauD and SsuD illustrates how gene duplication and functional divergence have produced substrate-specialized branches from a common architectural theme.

5.3 Ecological and physiological context

Taurine is one of the most abundant low-molecular-weight organic sulfur compounds in nature (notably abundant in animal tissues and bile), making it an ecologically important sulfur reservoir. The module is induced specifically when preferred sulfur sources (sulfate, cysteine) are exhausted, so its physiological "state" is sulfur starvation. Broader ecological work has highlighted taurine's role in host-microbiome interactions—for example, taurine-

utilizing taxa expand after infection and contribute to colonization resistance via sulfide production (PMID: 33453153)—though such systems-level phenomena involve catabolic (sulfide-producing) taurine metabolism rather than the sulfur-scavenging TauD route specifically and lie at the edge of this module's boundary.

6. Constraints, Dependencies, and Failure Modes

6.1 Ordered dependencies

Several steps are strictly ordered by compartmentalization and chemistry:

1. **Import before cleavage.** TauD is cytoplasmic; it can only act on taurine that TauABC has already imported. This is the defining dependency of the module—TauABC supplies intracellular taurine to TauD.
2. **Cofactor assembly order in TauD.** Fe(II) binds first, then 2OG chelates bidentately, then taurine binds, and only then does O₂ react. This sequential loading is what allows the enzyme to gate its dangerous oxidizing chemistry.
3. **2OG decarboxylation is coupled to ferryl formation.** Oxidative decarboxylation of 2OG to succinate + CO₂ generates the ferryl; the ferryl then abstracts H from taurine's C1.

6.2 Substrate gating and the self-hydroxylation failure mode

Finding F007. A defining constraint of TauD—and of the Fe/2OG superfamily generally—is that **O₂ activation must be gated by bound substrate**. When taurine is absent, the Fe(II)–2OG–TauD complex still reacts with O₂: 2OG is decarboxylated and a transient **Tyr73 radical** forms (an EPR-detectable yellow species, λ_{max} 408 nm), which decays to an Fe(III)-catecholate chromophore (λ_{max} 550 nm) arising from self-hydroxylation of Tyr73 to DOPA. Isotope labeling with ¹⁸O showed the added oxygen derives from solvent, not O₂ (PMID: 12590572): *"The transient yellow species, identified as a tyrosyl radical on the basis of EPR studies, is formed after alphaKG decomposition."*

Remarkably, TauD exhibits **more than one oxygen-activation route**. A distinct, succinate/H₂O₂-dependent self-hydroxylation produces a 720-nm species that interconverts with the 550-nm form via bound bicarbonate. Both converge on the same target: *"both reactions result in the self-hydroxylation of the active-site residue Tyr73"* (PMID: 16320009). The succinate-dependent reaction appears to use a different oxidant than the productive ferryl: *"we propose Fe(III)-OOH*

(or $\text{Fe(V)} = \text{O}$) as the oxygenating species in the succinate-dependent reaction" (PMID: 16320009). The DOPA-quinone form of Tyr73 can further react with Fe(II) to yield a semiquinone chromophore (PMID: 17973473). These self-hydroxylation reactions are unproductive, potentially self-inactivating side reactions that illustrate why substrate gating is essential: the enzyme's oxidizing power, if unleashed without a substrate to receive it, turns on the protein itself.

6.3 The ferryl mechanism and its second-sphere control

Finding F008. The productive chemistry runs through the ferryl intermediate abstracting a hydrogen atom from taurine's reactive C1. Direct proof comes from a **large substrate-deuterium kinetic isotope effect (KIE)**: deuteration at C1 produces "*an enormous kinetic isotope effect together with a partial uncoupling of oxygen activation from substrate oxidation*" (PMID: 19892731), directly implicating the ferryl in rate-limiting C–H bond cleavage.

The fidelity of coupling O_2 activation to substrate oxidation is controlled by **second-sphere residues**. Phe159, positioned directly behind the bound substrate, tunes this coupling: "*Decreasing side-chain bulk diminishes the coupling of oxygen activation to C–H cleavage*" (PMID: 19892731). Reducing Phe159's bulk (F159L/V/A/G) progressively uncouples the reaction, and even buffer (bis-Tris) concentration modulates the uncoupling pathway—yet O_2 activation remains fully coupled to 2OG decarboxylation. This shows a two-tier coupling logic: 2OG decarboxylation is tightly coupled to O_2 activation, but the use of the resulting ferryl for productive substrate hydroxylation depends on correct substrate positioning enforced by second-sphere residues.

Spectroelectrochemistry adds another layer: the Fe(II)/Fe(III) redox potential of TauD spans ~468 mV with a large redox-linked conformational reorganization dependent on the facial-triad residues His99/Asp101/His255 (PMID: 31475523).

6.4 Evidence ruling out alternative paths

- The strict O_2 dependence and requirement for 2OG rule out the anaerobic transaminase route within the TauD system—these are separate enzymes in separate organisms/conditions.
- The non-functionality of tau/ssu hybrid transporters (PMID: 10781534) rules out free interchange of importer components.
- The large C1-KIE (PMID: 19892731) rules out mechanisms in which C–H cleavage is not rate-limiting, and localizes the abstracted hydrogen to C1 specifically.

- ^{18}O labeling showing solvent (not O_2) as the source of the self-hydroxylation oxygen ([PMID: 12590572](#)) constrains the mechanism of the off-pathway reaction.

7. Controversies and Open Questions

7.1 Rebound versus alkoxide: how does hydroxylation finish?

The best-supported claim is that the ferryl abstracts a hydrogen atom from taurine C1 (established by the large KIE). What remains genuinely unresolved is **how the substrate radical is converted to the 1-hydroxylated product**. As stated in the spectroelectrochemistry study, "*Prior studies raised the question of whether substrate hydroxylation by these enzymes occurs via a hydroxyl rebound or alkoxide mechanism*" ([PMID: 31475523](#)). In the classic **oxygen-rebound** picture, the substrate radical recombines with the Fe–OH to install the hydroxyl; in the **alkoxide** alternative, the pathway proceeds differently. TauD is the reference system in which this mechanistic question is being adjudicated, and it is not yet settled.

7.2 Structural characterization of the reactive ferryl

Because the ferryl is short-lived, it has resisted crystallography. Recent work uses **vanadyl (V=O) substitution** as a stable structural mimic of the reactive ferryl, yielding crystal structures that approximate intermediate J ([PMID: 31503454](#); [PMID: 28960972](#)). How faithfully these mimics reproduce the true ferryl geometry—particularly the orientation of the Fe–O bond relative to the target C–H bond, which governs reactivity and selectivity—remains an active question.

7.3 Multiple oxidants in one active site

The observation that TauD supports at least two distinct oxygen-activation chemistries (the productive ferryl and a succinate/ H_2O_2 -dependent Fe(III)-OOH/Fe(V)=O oxidant) raises the question of how general such promiscuity is across the superfamily and whether it has physiological relevance or is merely an in-vitro artifact of substrate-free conditions ([PMID: 16320009](#)).

7.4 Transport mechanism and structural biology of TauABC

Compared with the exhaustive mechanistic dissection of TauD, the **TauABC transporter is far less structurally characterized**. The desolvation-based selectivity of TauA rests substantially on MD simulations ([PMID: 31802112](#)); a full-length transporter structure and direct measurement of transport kinetics and coupling stoichiometry are lacking. How the periplasmic binding protein hands off taurine to the TauBC complex, and the conformational cycle of translocation, remain to be resolved.

7.5 Cross-organism comparability

Nearly all detailed mechanistic data come from *E. coli* (and *P. putida* for oligomeric state and some kinetics). Regulatory detail (CysB/Cbl/APS) is *E. coli*-centric, and the anaerobic routes come from entirely different organisms. Readers should be cautious about assuming the *E. coli* module's precise regulation, kinetics, and structure generalize across all taurine-utilizing bacteria.

7.6 The most important open questions

1. Rebound vs alkoxide mechanism of the final hydroxylation step (§7.1).
2. A bona fide structure of the reactive ferryl (vs vanadyl mimics) (§7.2).
3. Structural and kinetic characterization of the TauABC transport cycle (§7.4).
4. The generality and physiological relevance of TauD's alternative oxidants (§7.3).
5. How broadly the *E. coli* regulatory and mechanistic model applies across lineages (§7.5).

8. Key References

The following are the primary papers supporting the findings in this review, with the specific role each plays.

PMID	Title (abbrev.)	Supports
8808933	Identification of sulfate starvation-regulated genes; tau gene cluster	Defines the two-part module (F001)
10781534	Deletion analysis of E. coli taurine/ alkanesulfonate transport	Conditional expression; non-functional hybrids (F001, F002)
31802112	Desolvation of TauA dictates ligand specificity	Desolvation-based selectivity of TauA (F002)
12809506	First direct characterization of a high-valent Fe intermediate in TauD	Ferryl "intermediate J" (F003)
22221834	TauD from P. putida and E. coli are tetramers	Products, cofactor, oligomeric state (F003)
15751960	Steady-state/transient kinetics of TauD	Active-site substrate recognition (F003)
11955067	X-ray crystal structure of E. coli TauD	DSBH fold, facial triad, substrate geometry (F006)
17350690	Metal ligand substitution in TauD	Facial-triad ligand mutagenesis (F006)
11479697	Sulfonate-sulfur metabolism and regulation in E. coli	CysB/Cbl regulation (F004)
11918818	APS as signaling molecule for sulfate excess	APS as the repressive signal (F004)
10717312	Metabolism of sulfonates in gram-negative bacteria	2OG vs FMNH ₂ routes (F005)
11728723	R. capsulatus taurine sulfur-source gene region	Anaerobic Tpa route (F005)
12590572	O ₂ /2OG-dependent tyrosyl radical formation in TauD	Tyr73 radical / uncoupling (F007)
16320009	Self-hydroxylation of TauD; >1 oxygen activation mechanism	Tyr73 self-hydroxylation; multiple oxidants (F007)
19892731	Modular behavior of TauD / origin of specificity	Large KIE; Phe159 coupling control (F008)
31475523	Redox-linked reorganization in TauD	Redox potential; rebound-vs-alkoxide question (F008)

PMID	Title (abbrev.)	Supports
31503454	Structure of a ferryl mimic (vanadyl) in TauD	Ferryl structural mimic (§7.2)
28960972	Vanadyl as stable mimic of ferryl intermediates	Ferryl structural mimic (§7.2)
17973473	Cr(II) reactivity of TauD	DOPA-quinone/semiquinone chemistry (F007)

Limitations of This Review

This synthesis is built from a focused literature base of ~27 papers and is strongly weighted toward *E. coli* and, secondarily, *P. putida*. The mechanistic depth on TauD greatly exceeds that on TauABC, so the review is necessarily more detailed and more confident about desulfonation than about transport. Regulatory conclusions (CysB/Cbl/APS) are *E. coli*-specific. Where claims rest on indirect evidence—MD simulations for TauA selectivity, vanadyl mimics for the ferryl, and unresolved rebound-vs-alkoxide chemistry—this has been stated explicitly. The anaerobic catabolic routes are included only to delineate the module's boundary and are not reviewed in comparable depth.

Proposed Follow-up Directions

1. **Solve a full-length TauABC transporter structure** (cryo-EM) with and without bound taurine to define the translocation cycle and test the desolvation-selectivity model directly.
2. **Adjudicate the rebound-vs-alkoxide question** in TauD using time-resolved crystallography, computational QM/MM, and isotope-tracking of the installed hydroxyl.
3. **Validate vanadyl ferryl mimics** against the true intermediate by comparing predicted and observed reactivity/selectivity as a function of Fe–O/C–H geometry.
4. **Map the regulatory circuit across lineages** to test how broadly the APS/Cbl/CysB logic applies beyond *E. coli*.
5. **Quantify in-vivo flux** through TauABC→TauD under graded sulfur limitation to establish which step is rate-controlling for sulfur acquisition and whether self-hydroxylation ever limits productivity physiologically.

Prepared as a commissioned review of the bacterial taurine uptake and desulfonation module (TauABC import → TauD desulfonation). All mechanistic claims are anchored to the cited primary literature; uncertainty is flagged where the evidence is indirect or organism-specific.

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